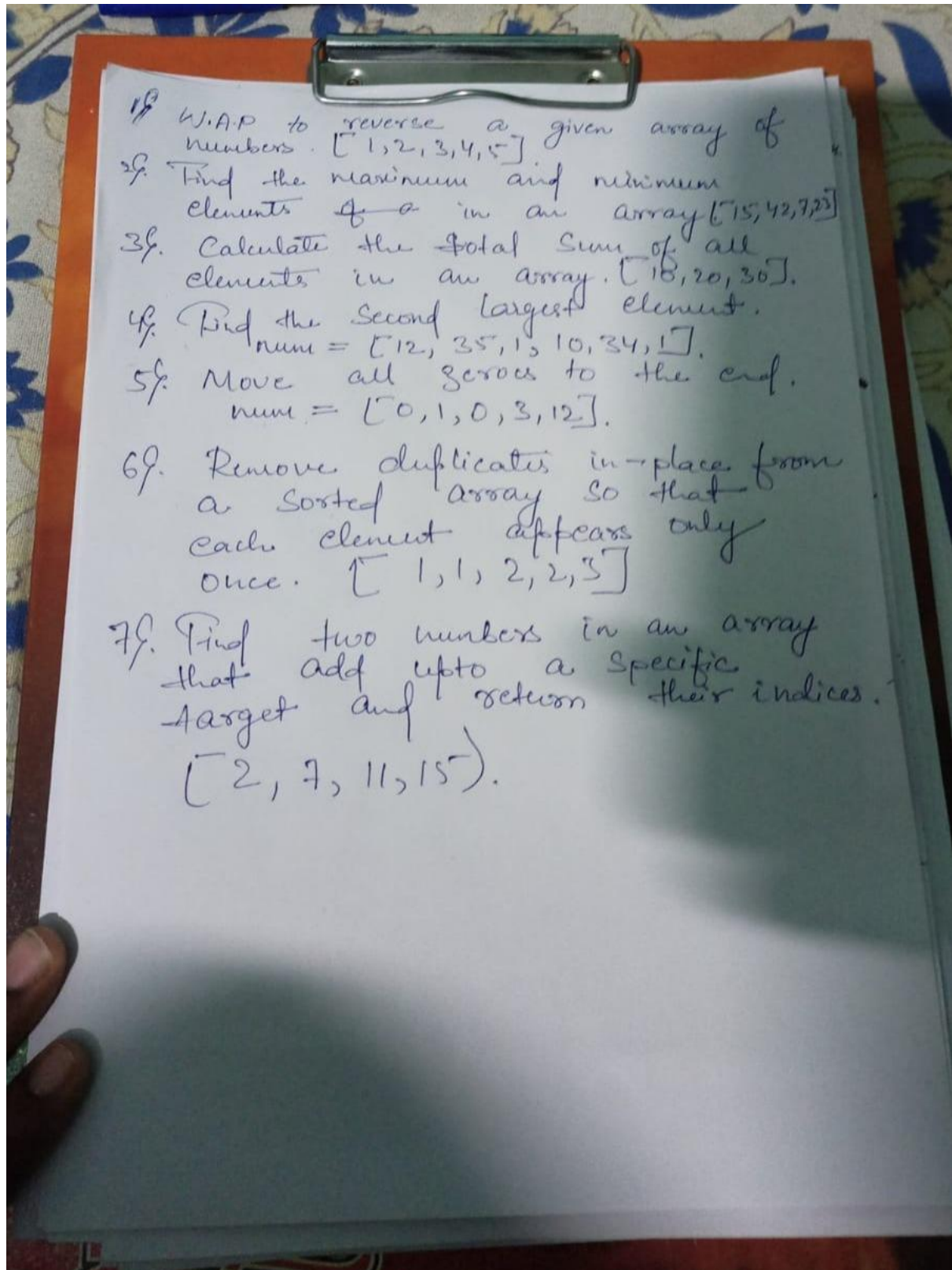




Q1.

```
57_array01.py > ...
1  import array as arr
2
3  array1 = arr.array('i',[1,2,3,4,5])
4
5  print(array1)
6
7  rev_array = array1[::-1]
8
9  print(rev_array)
10
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PO

```
● PS D:\ML\assignments> & "C:\Program Files\Python314
array('i', [1, 2, 3, 4, 5])
array('i', [5, 4, 3, 2, 1])
○ PS D:\ML\assignments> |
```

Q2.

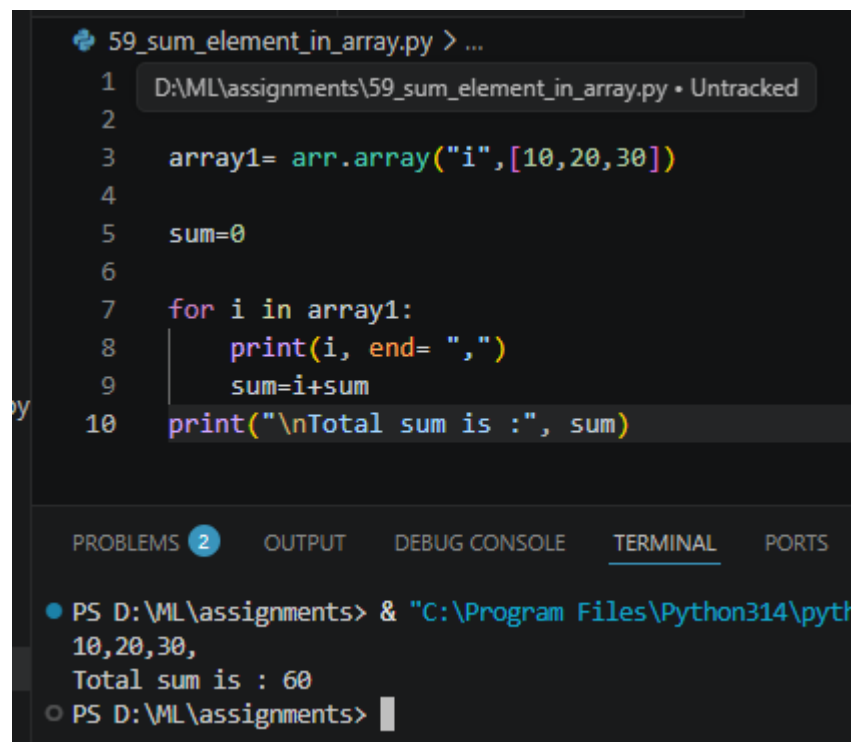
```
58_array_lowest_greatest.py > ...
1  #15,42,7,23
2
3  import array as arr
4  array2 = arr.array('i',[15,42,7,23])
5
6  print(array2)
7
8  great=0
9  lowest=999
10 for i in array2:
11     print(i,"\n\n")
12     for j in array2:
13
14         if j == i:
15             continue
16         elif j>great:
17             great = j
18         elif j<lowest:
19             lowest=j
20         print(j,end= ",")
21     print("\n\n")
22 print("greatest no. : ", great)
23 print("lowest no. : ",lowest)
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\ML\assignments> & "C:\Program Files\Python314\python.exe" d:/M

greatest no. : 42
lowest no. : 7
PS D:\ML\assignments>
```

Q3.



The image shows a code editor window with a file named `59_sum_element_in_array.py`. The code defines an array `array1` with values `[10, 20, 30]`, initializes a `sum` variable to 0, and uses a `for` loop to iterate through the array, printing each element and accumulating the sum. The final output is `Total sum is : 60`.

```
1  D:\ML\assignments\59_sum_element_in_array.py • Untracked
2
3  array1= arr.array("i",[10,20,30])
4
5  sum=0
6
7  for i in array1:
8      print(i, end= ",")
9      sum=i+sum
10 print("\nTotal sum is :", sum)
```

Below the code editor, the **TERMINAL** tab is active, showing the command prompt output:

```
● PS D:\ML\assignments> & "C:\Program Files\Python314\pyth
10,20,30,
Total sum is : 60
○ PS D:\ML\assignments> |
```

Q4.

```
60_second_greatest.py > ...
1  #12,35,1,10,34,1
2  import array as arr
3  arr = arr.array("i",[12,35,1,10,34,1])
4  print(arr)
5  great=0
6  great_list=[]
7  for i in arr:
8      for j in arr:
9          if j==i:
10             continue
11             elif i>great:
12                 great=i
13             else:
14                 continue
15
16  print(great)
17  arr.remove(great)
18  print(arr)
19
20  s_great=0
21  for i in arr:
22      for j in arr:
23          if j==i:
24             continue
25             elif j>i:
26                 s_great=j
27             else:
28                 continue
29  print("Second greatest no in array is :",s_great)
30
```

OUTPUT:

```
PS D:\ML\assignments> & "C:\Program Files\Python\Python38\python.exe" 60_second_greatest.py
array('i', [12, 35, 1, 10, 34, 1])
35
array('i', [12, 1, 10, 34, 1])
Second greatest no in array is : 34
PS D:\ML\assignments>
```

Q5.

```
63_array_movetoend.py > ...
1  import array as arr
2
3  array1 = arr.array("i",[0,1,0,3,12])
4
5  print(array1)
6  for i in array1:
7      print(i)
8      if i==0:
9          array1.remove(i)
10         array1.append(i)
11 print(array1)
12
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\ML\assignments> & "C:\Program Files\Python314\python.exe"
0
0
12
0
0
array('i', [1, 3, 12, 0, 0])
PS D:\ML\assignments>
```

Q.6

```
64_array_remove_duplicate_sorted.py > ...
28
29 import array as arr
30
31 def remove_duplicate(nums):
32     if not nums:
33         return 0
34     i=0
35     for j in range(1,len(nums)):
36         if nums[j] != nums[i]:
37             i += 1
38             nums[i] = nums[j]
39     return i + 1
40 ar = arr.array("i",[1,1,2,2,3])
41 k = remove_duplicate(ar)
42
43 print(f"no. of ununique elements: {k}")
44 print(f"Modified array: {ar[:k]}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS D:\ML\assignments> & "C:\Program Files\Python314\python.exe" d:/M
no. of ununique elements: 3
Modified array: array('i', [1, 2, 3])
○ PS D:\ML\assignments> 
```

Q7

```
65_TWOSUM_PROBLEM.PY > ...
1  def two_sum(nums, target):
2      # Dictionary to store the number and its index
3      num_map = {}
4
5      for i, num in enumerate(nums):
6          # Calculate what number we need to hit the target
7          complement = target - num
8
9          # If the complement is in our map, we found our pair
10         if complement in num_map:
11             return [num_map[complement], i]
12
13         # Otherwise, add the current number and its index to the map
14         num_map[num] = i
15
16     return [] # Return empty if no pair is found
17
18 # Test the function (assuming a target of 9)
19 arr = [2, 7, 11, 15]
20 target_val = 9
21 indices = two_sum(arr, target_val)
22
23 print(f"Indices for target {target_val}: {indices}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORT

```
PS D:\ML\assignments> & "C:\Program Files\Python310\python.exe" 65_TWOSUM_PROBLEM.PY
Indices for target 9: [0, 1]
PS D:\ML\assignments>
```